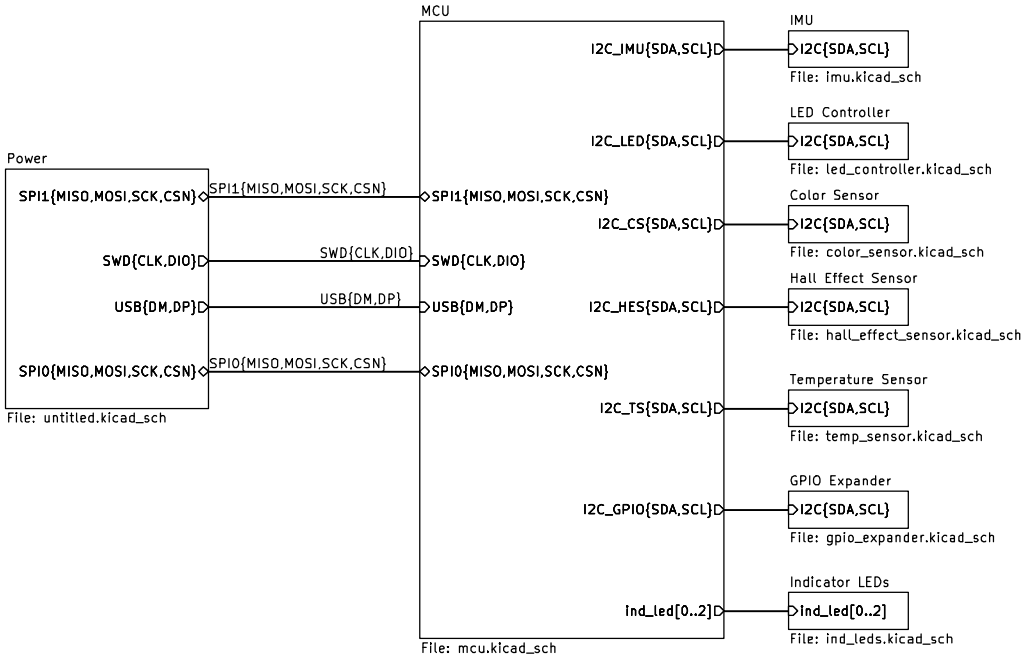
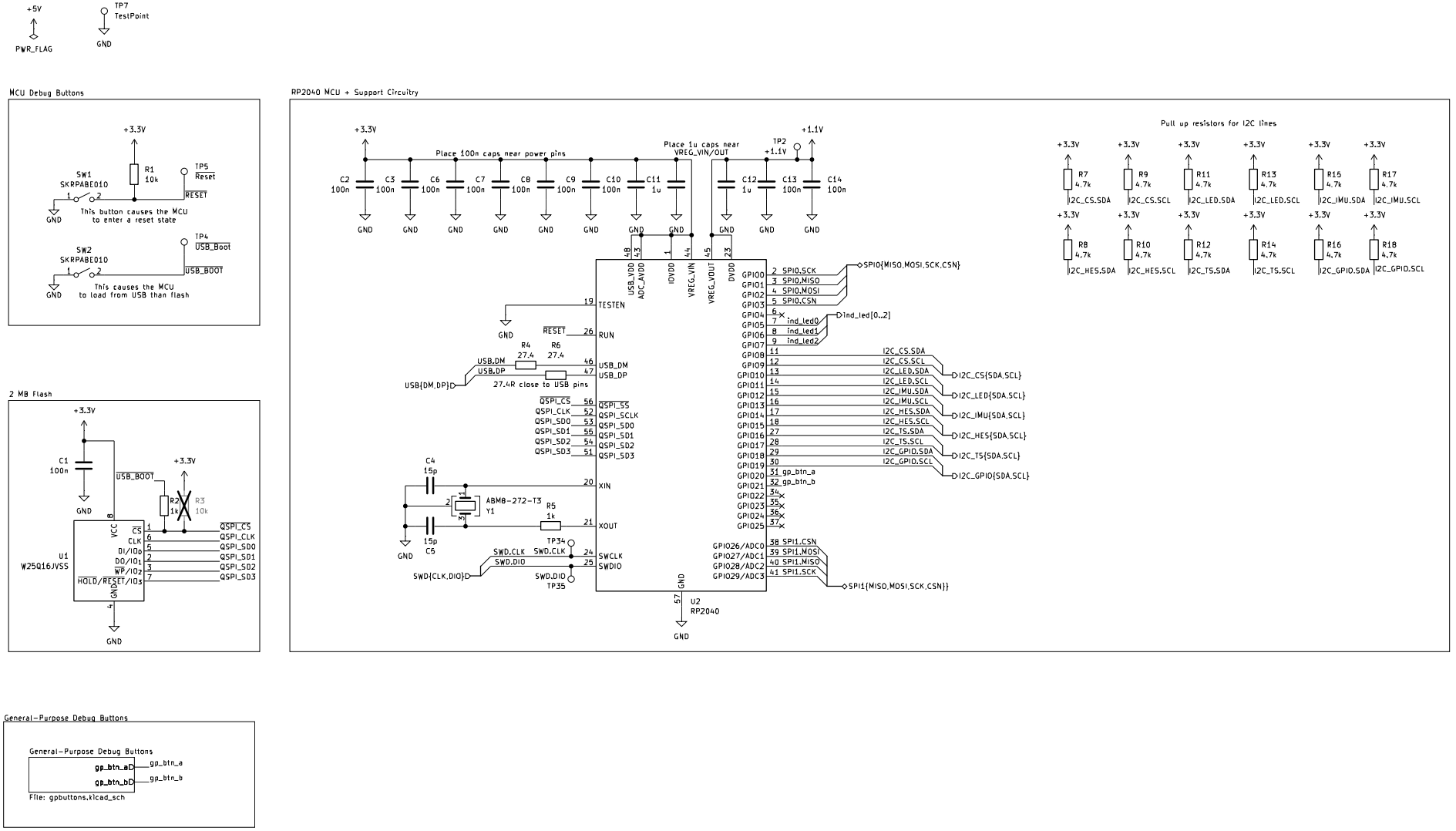
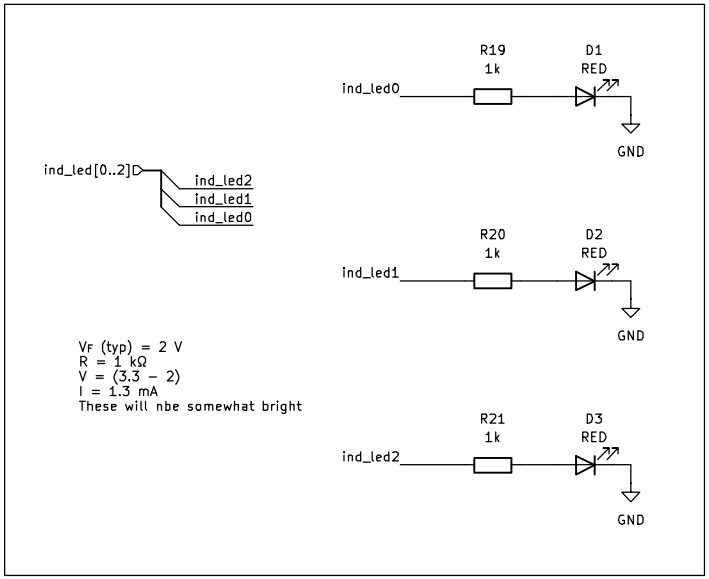
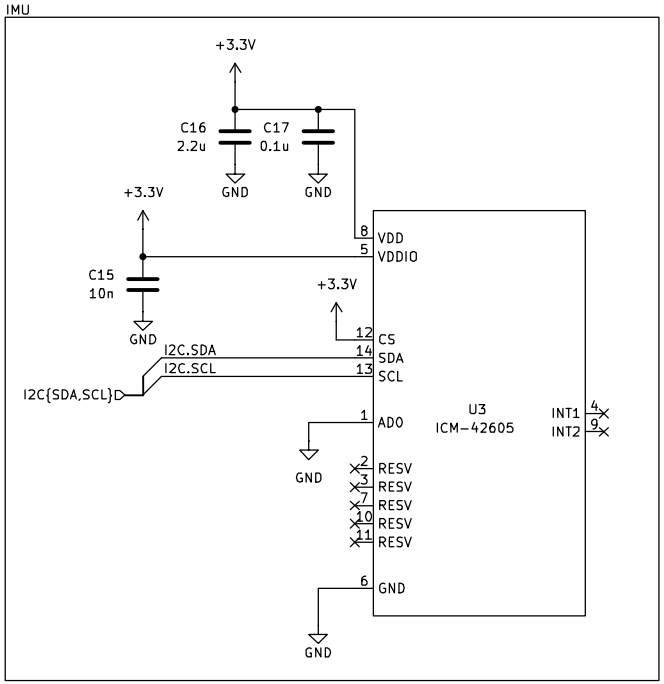

Power
SPI1{MISO,MOSI,SCK,CSN}
SWD{CLK,DIO}
USB{DM,DP}
SPI0{MISO,MOSI,SCK,CSN}
File: untitled.kicad_sch
SPI1{MISO,MOSI,SCK,CSN}
SWD{CLK,DIO}
USB{DM,DP}
SPI0{MISO,MOSI,SCK,CSN}
MCU
SPI1{MISO,MOSI,SCK,CSN}
SWD{CLK,DIO}
USB{DM,DP}
SPI0{MISO,MOSI,SCK,CSN}
I2C_IMU{SDA,SCL}
I2C_LED{SDA,SCL}
I2C_CS{SDA,SCL}
I2C_HES{SDA,SCL}
I2C_TS{SDA,SCL}
I2C_GPIO{SDA,SCL}
Ind_led[0..2]
File: mcu.kicad_sch
IMU
I2C{SDA,SCL}
File: imu.kicad_sch
LED Controller
I2C{SDA,SCL}
File: led_controller.kicad_sch
Color Sensor
I2C{SDA,SCL}
File: color_sensor.kicad_sch
Hall Effect Sensor
I2C{SDA,SCL}
File: hall_effect_sensor.kicad_sch
Temperature Sensor
I2C{SDA,SCL}
File: temp_sensor.kicad_sch
GPIO Expander
I2C{SDA,SCL}
File: gpio_expander.kicad_sch
Indicator LEDs
Ind_led[0..2]
File: ind_leds.kicad_sch


+5V
PWR_FLAG
TP7
TestPoint
GND
MCU Debug Buttons
+3.3V
R1
10k
SW1
SKRPABE010
TP5
Reset
RESET
GND
This button causes the MCU
to enter a reset state
TP4
USB_Boot
SW2
SKRPABE010
USB_BOOT
This causes the MCU
to load from USB than flash
2 MB Flash
C1
100n
USB_BOOT
R2
1k
R3
10k
U1
W25Q16JVSS
VCC
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
GND
QSPI_CS
QSPI_CLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
RP2040 MCU + Support Circuitry
Place 100n caps near power pins
Place 1u caps near
VREG_VIN/VOUT
C2
100n
C3
100n
C6
100n
C7
100n
C8
100n
C9
100n
C10
100n
C11
1u
C12
1u
C13
100n
C14
100n
TP2
+1.1V
+1.1V
USB_VDD
ADC_AVDD
IOVDD
VREG_VIN
VREG_VOUT
DVDD
TESTEN
RUN
USB_DM
USB_DP
R4
27.4
R6
27.4
USB{DM,DP}
27.4R close to USB pins
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
C4
15p
C5
15p
Y1
ABM8-272-T3
R5
1k
XIN
XOUT
SWCLK
SWDIO
SWD.CLK
SWD.DIO
SWD{CLK,DIO}
TP34
TP35
U2
RP2040
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26/ADC0
GPIO27/ADC1
GPIO28/ADC2
GPIO29/ADC3
SPI0.SCK
SPI0.MISO
SPI0.MOSI
SPI0.CSN
SPI0{MISO,MOSI,SCK,CSN}
ind_led0
ind_led1
ind_led2
ind_led[0..2]
gp_btn_a
gp_btn_b
SPI1.CSN
SPI1.MOSI
SPI1.MISO
SPI1.SCK
SPI1{MISO,MOSI,SCK,CSN}
I2C_CS.SDA
I2C_CS.SCL
I2C_LED.SDA
I2C_LED.SCL
I2C_IMU.SDA
I2C_IMU.SCL
I2C_HES.SDA
I2C_HES.SCL
I2C_TS.SDA
I2C_TS.SCL
I2C_GPIO.SDA
I2C_GPIO.SCL
I2C_CS{SDA,SCL}
I2C_LED{SDA,SCL}
I2C_IMU{SDA,SCL}
I2C_HES{SDA,SCL}
I2C_TS{SDA,SCL}
I2C_GPIO{SDA,SCL}
Pull up resistors for I2C lines
R7
4.7k
R8
4.7k
R9
4.7k
R10
4.7k
R11
4.7k
R12
4.7k
R13
4.7k
R14
4.7k
R15
4.7k
R16
4.7k
R17
4.7k
R18
4.7k
General-Purpose Debug Buttons
General-Purpose Debug Buttons
gp_btn_a
gp_btn_b
File: gpbuttons.kicad_sch

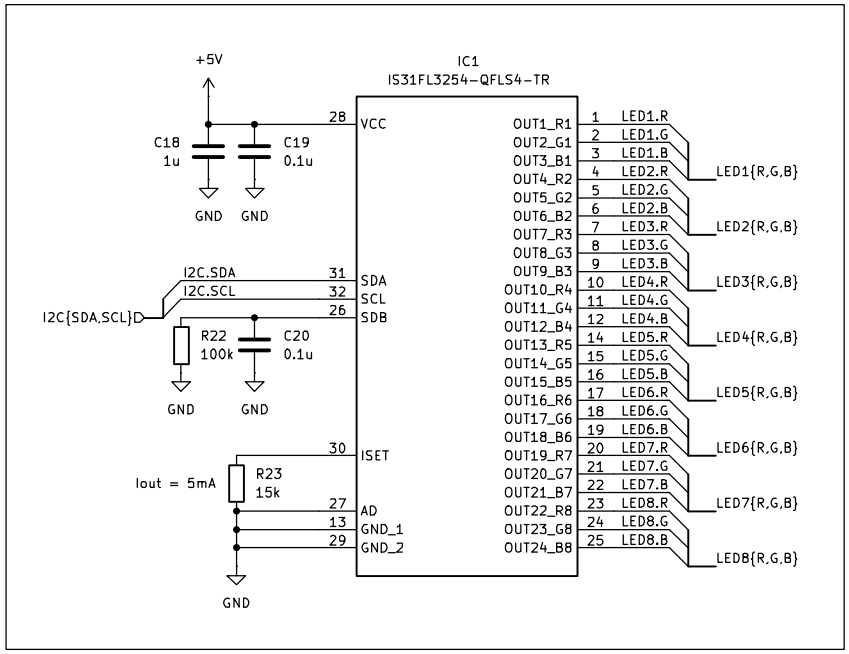
Indicator LEDs





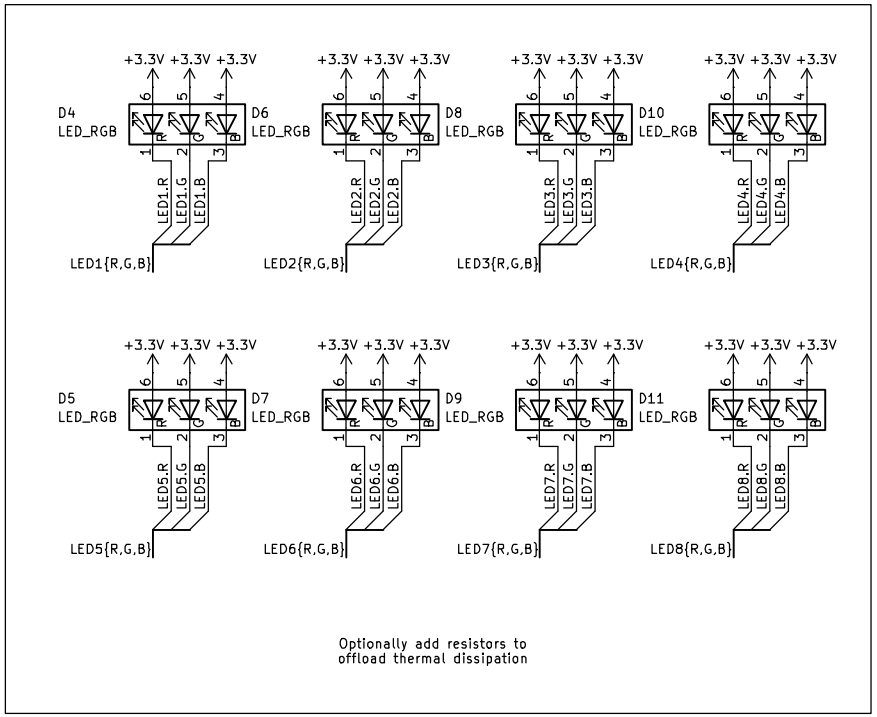

| | |
|---|---|
| Device | ICM-42605 (U8) |
| Function | Interial Measurement Unit |
| I2C Address (7-bit) | 1101000 (0x68) |
| Address Config | AD0 on GND |
| Voltage Range | 1.7-3.6V |

LED Driver

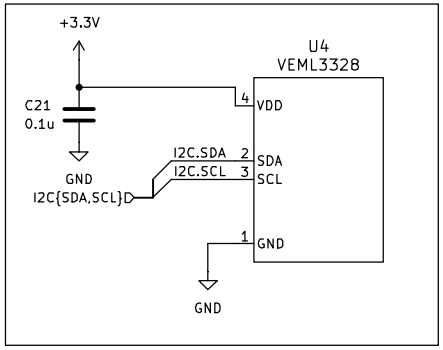



| | |
|---|---|
| Device | IS31FL3254-QFLS4-TR (IC1) |
| Function | LED Controller |
| I2C Address (7-bit) | 0110000 (0x30) |
| Address Config | AD on GND |
| Voltage Range | 2.7-5.5V |
| Pull-Ups | 4.7k on I2C |
| Led 0 Pin | RP2040 GPIO 5 (Pin 7) |
| Led 1 Pin | RP2040 GPIO 6 (Pin 8) |
| Led 2 Pin | RP2040 GPIO 7 (Pin 9) |

RGB LEDs



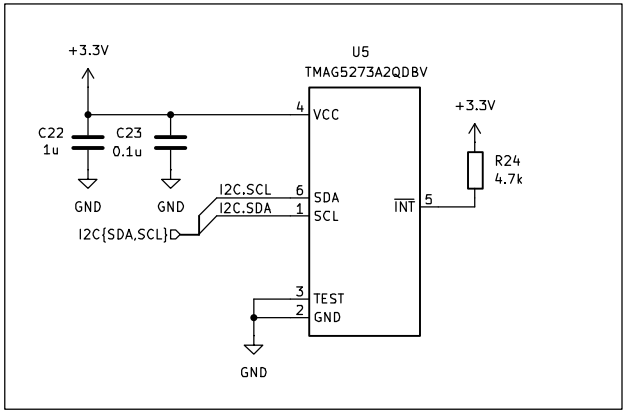
Color Sensor




| | |
|---|---|
| Device | VEML3328 (U4) |
| Function | Color Sensor |
| I2C Address (7-bit) | 0101000 (0x28) |
| Address Config | NONE |
| Voltage Range | 2.6-3.6V |

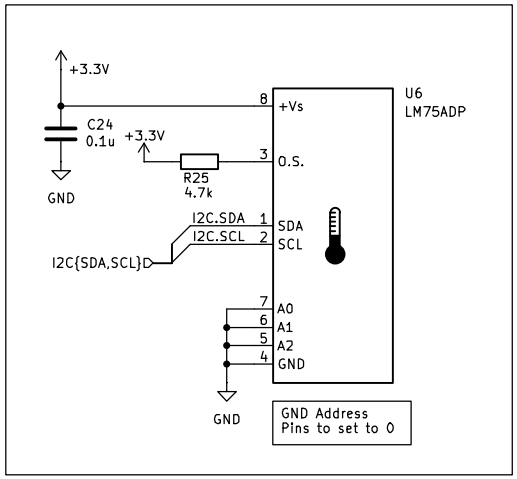
| | |
|---|---|
| Device | TMAG5273A2 (U5) |
| Function | Hall Effect & Temperature Sensor |
| I2C Address (7-bit) | 0110101 (0x35) |
| Address Config | Register 0x0C |
| Voltage Range | 1.7–3.6V |
| Pull-Ups | 4.7k on INT and I2C |

Hall Effect Sensor



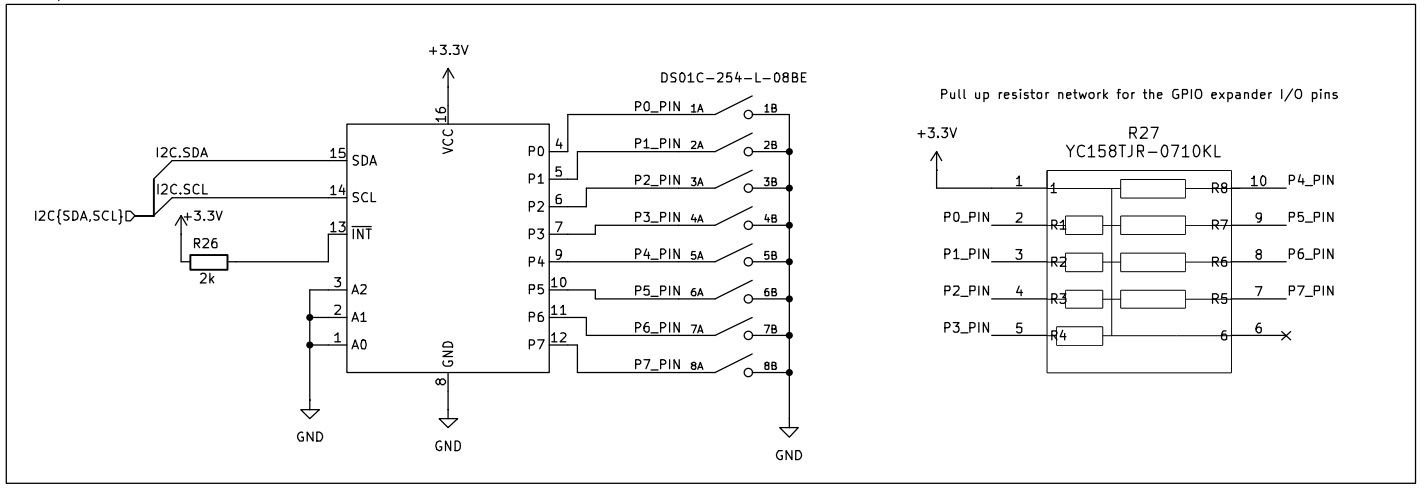

Datasheet advises to ground TEST

Temperature Sensor



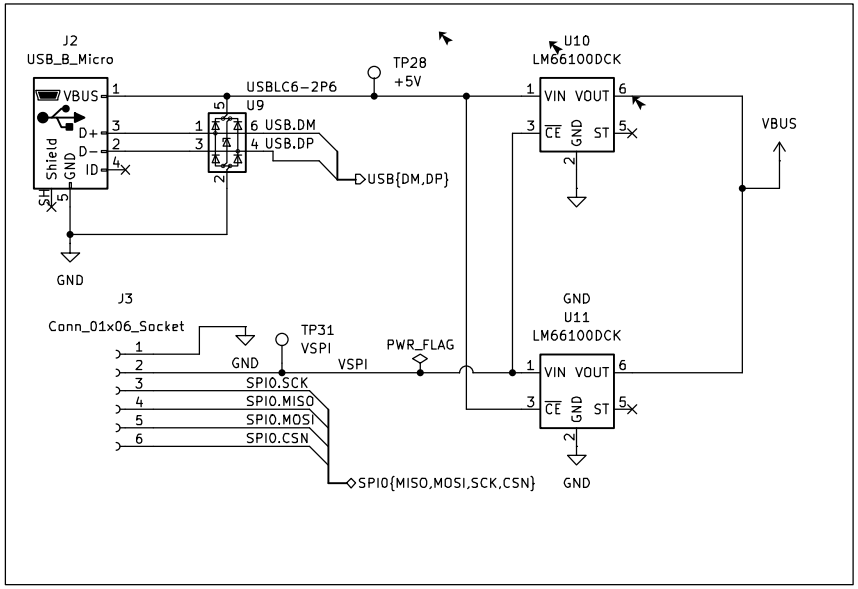

| | |
|---|---|
| Device | LM75ADP (U6) |
| Function | Temperature Sensor |
| I2C Address (7-bit) | 1001000 (0x48) |
| Address Config | A0-A2 on GND |
| Voltage Range | 2.8-5.5V |
| Pull-Ups | 4.7k on OS and I2C |

GPIO Expander

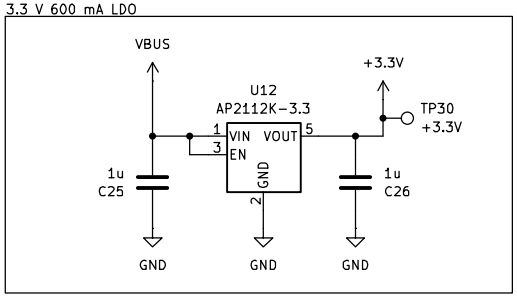



| | |
|---|---|
| Device | TCA9534PWR (U7) |
| Function | GPIO Expander |
| I2C Address (7-bit) | 0100000 (0x20) |
| Address Config | A0-A2 on GND |
| Voltage Range | 1.65-5.5V |
| Pull-Ups | 4.7k on I2C, 2K on INT to VCC |

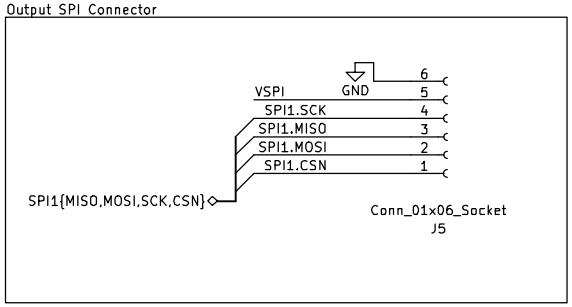
USB-C and SPI Input Power MUXing



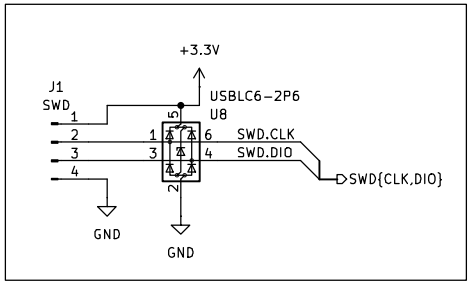

3.3 V 600 mA LDO




Output SPI Connector




SWD Debug Connector



Debug and Option Buttons

